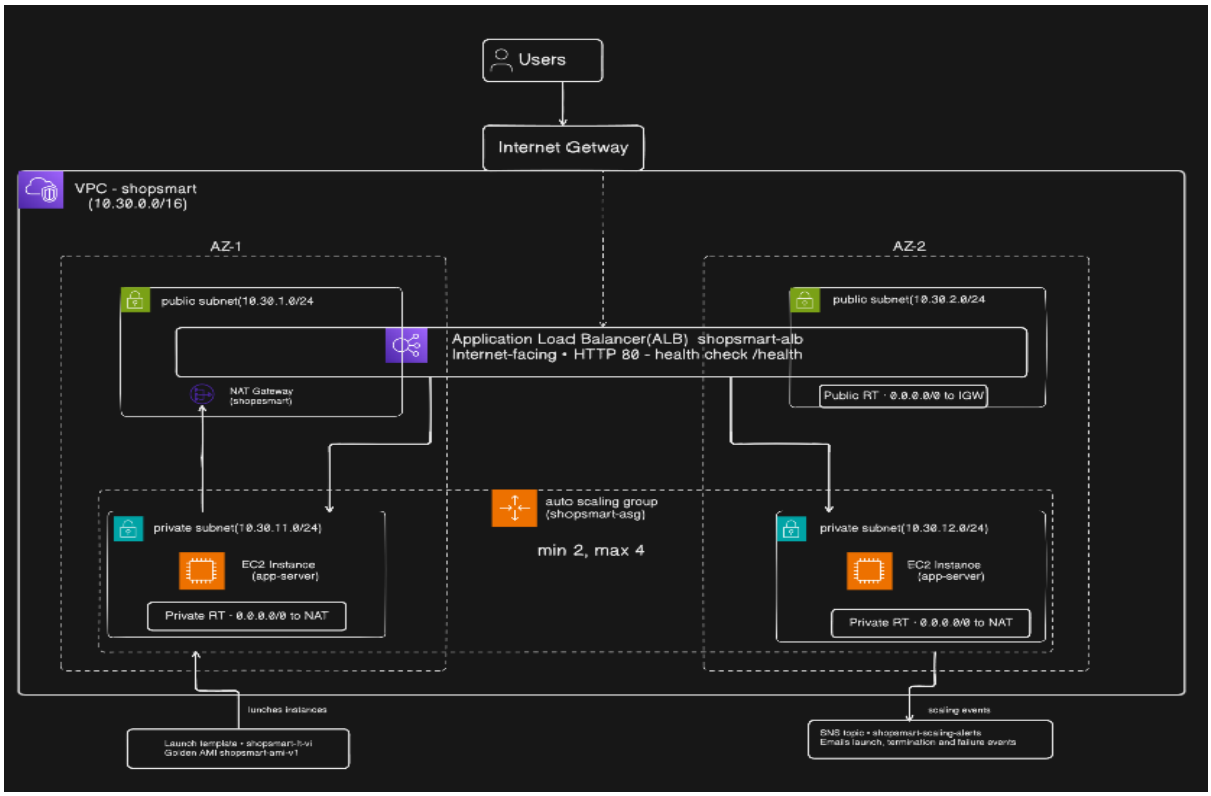
Production-Grade AWS EC2 Project: Highly Available Web

Application Project name: ShopSmart — Auto-Scaling E-Commerce Web Infrastructure

1: An architecture diagram showing the ALB, EC2 instances, Availability Zones, subnets, and Auto Scaling Group.

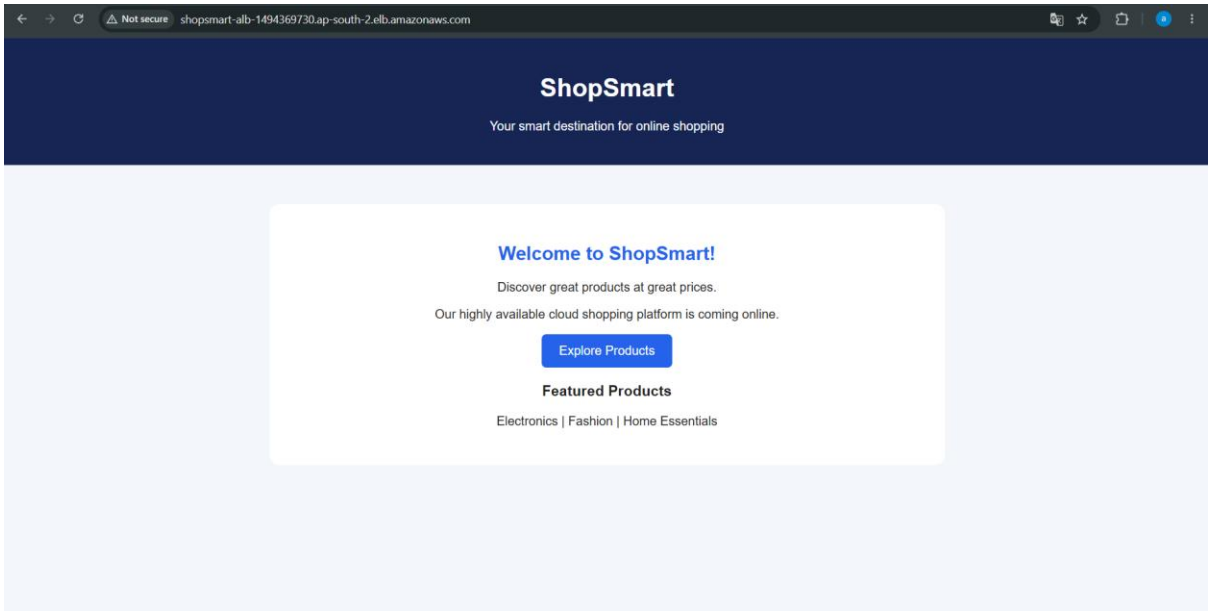


Customers reach the internet-facing Application Load Balancer through the Internet gateway. The ALB sits in two public subnets (10.30.1.0/24 and 10.30.2.0/24) across two Availability Zones and forwards traffic only to healthy EC2 instances. The instances run in private subnets (10.30.11.0/24 and 10.30.12.0/24) and are managed by an Auto Scaling Group (min 2, desired 2, max 4) that replaces failed instances and scales on CPU. The group launches from a launch template that uses the golden AMI. Private instances reach the internet

for updates through the NAT Gateway, and scaling events are sent to an SNS topic.

2: A screenshot of the working application through the ALB DNS name:

The ShopSmart web application was successfully accessed through the DNS name of the internet-facing Application Load Balancer. The ALB receives HTTP requests and forwards them to healthy registered EC2 instances running the application in private subnets.

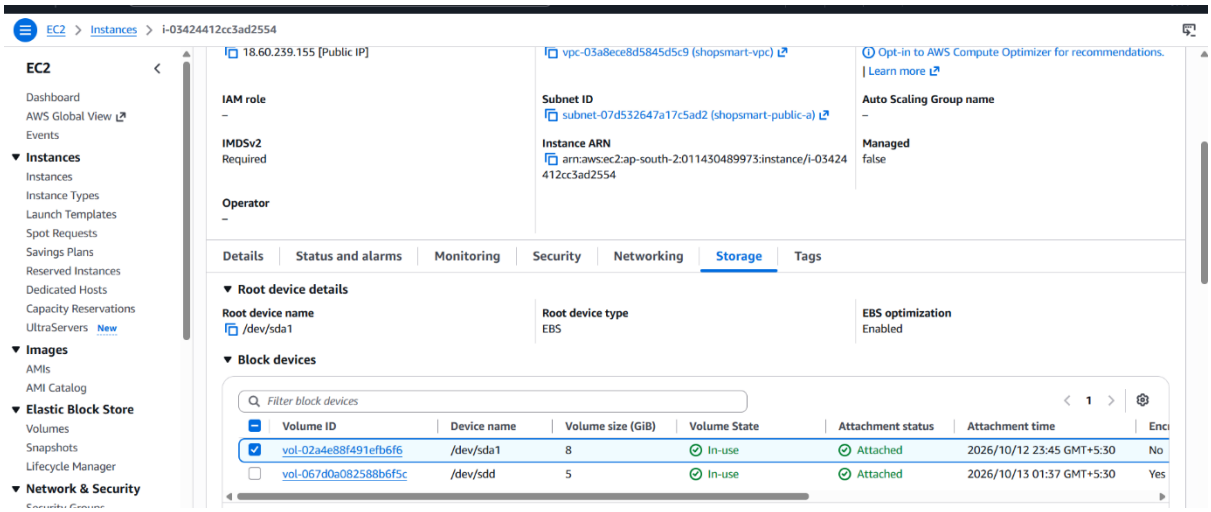


3. EBS Volume Attachment, Snapshot Creation, and AMI-Based Instance Creation

3.1 EBS Volume Attachment

The ShopSmart EC2 instance uses an 8 GiB root EBS volume and an additional 5 GiB EBS volume. The additional volume is mounted at /data in the operating system. The AWS Console confirms that both

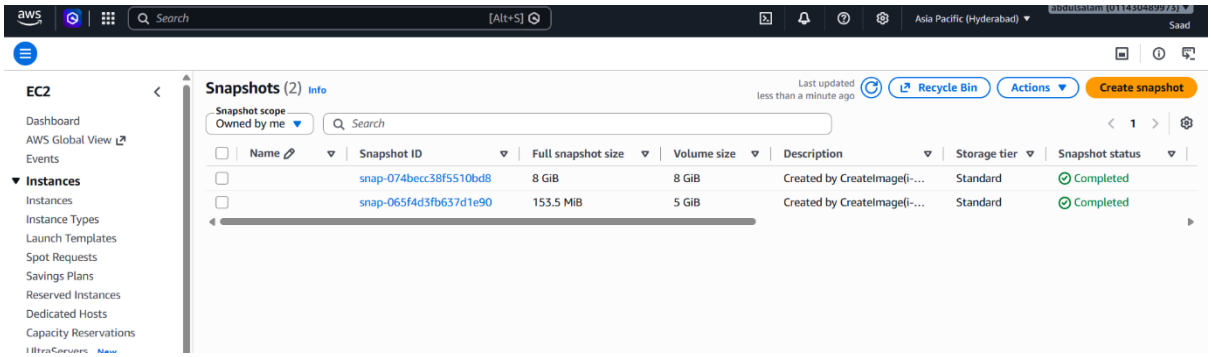
volumes are attached to the instance.




```
ubuntu@ip-10-30-1-42:~$ sudo mkdir -p /data
ubuntu@ip-10-30-1-42:~$ ls /
bin   data  etc   lib    lost+found  mnt  proc  run   snap  sys  usr
boot  dev   home  lib64  media       opt  root  sbin  srv   tmp  var
ubuntu@ip-10-30-1-42:~$ sudo mount /dev/nvme1n1 /data
ubuntu@ip-10-30-1-42:~$ lsblk
NAME         MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0          7:0    0 28.2M  1 loop /snap/amazon-ssm-agent/13009
loop1          7:1    0 49.3M  1 loop /snap/snapd/26865
loop2          7:2    0   74M  1 loop /snap/core22/2411
nvme0n1      259:0    0    8G  0 disk
├─nvme0n1p1  259:1    0  6.9G  0 part /
├─nvme0n1p13 259:2    0 1023M  0 part /boot
├─nvme0n1p14 259:3    0    4M  0 part
└─nvme0n1p15 259:4    0  106M  0 part /boot/efi
nvme1n1      259:5    0    5G  0 disk /data
```

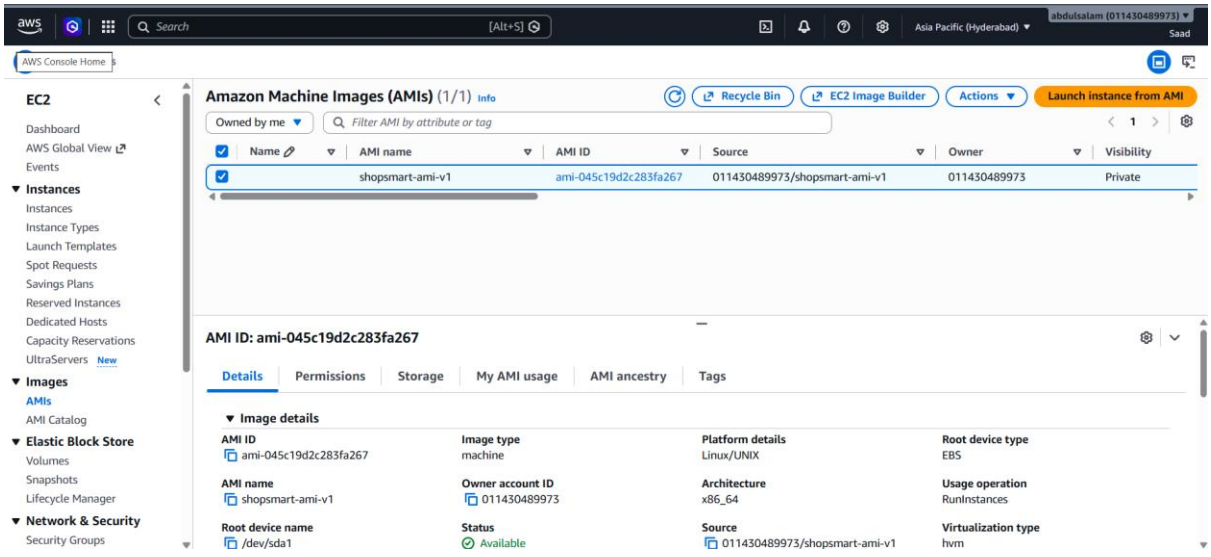
3.2 EBS Snapshot Creation

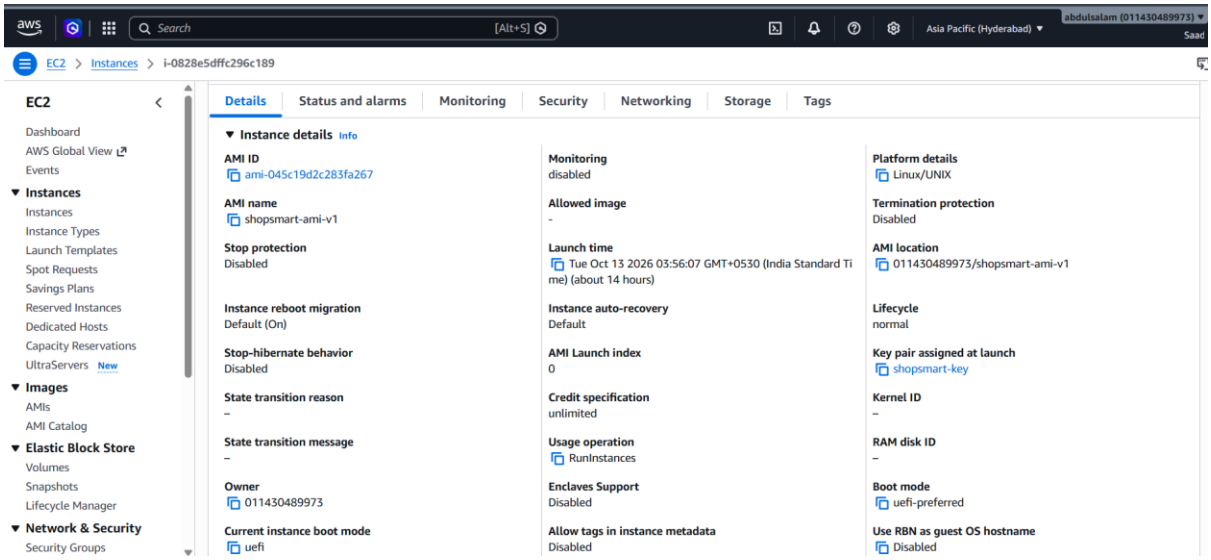
Two EBS snapshots were created successfully. The snapshots have volume sizes of 8 GiB and 5 GiB, respectively, and both show the status *Completed*. These snapshots provide recovery points for the associated EBS storage.



3.3 Amazon Machine Image (AMI)

A custom AMI named shopsmart-ami-v1 was created and is available for launching EC2 instances. The AMI contains the prepared Ubuntu server environment and ShopSmart landing page configuration. It can be used as the image source for launching additional instances with a consistent configuration

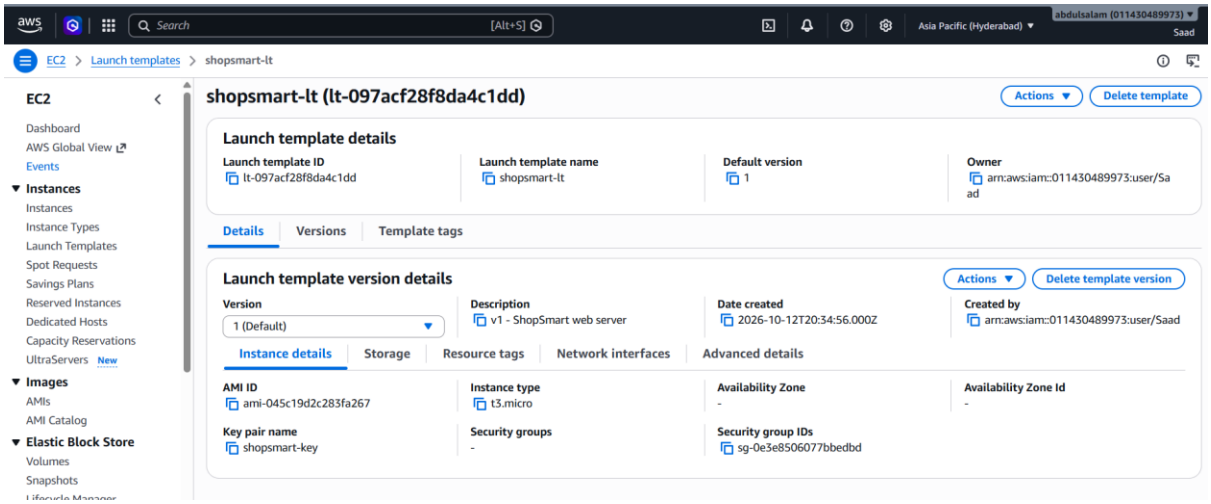




4: The launch template configuration and Auto Scaling policy.

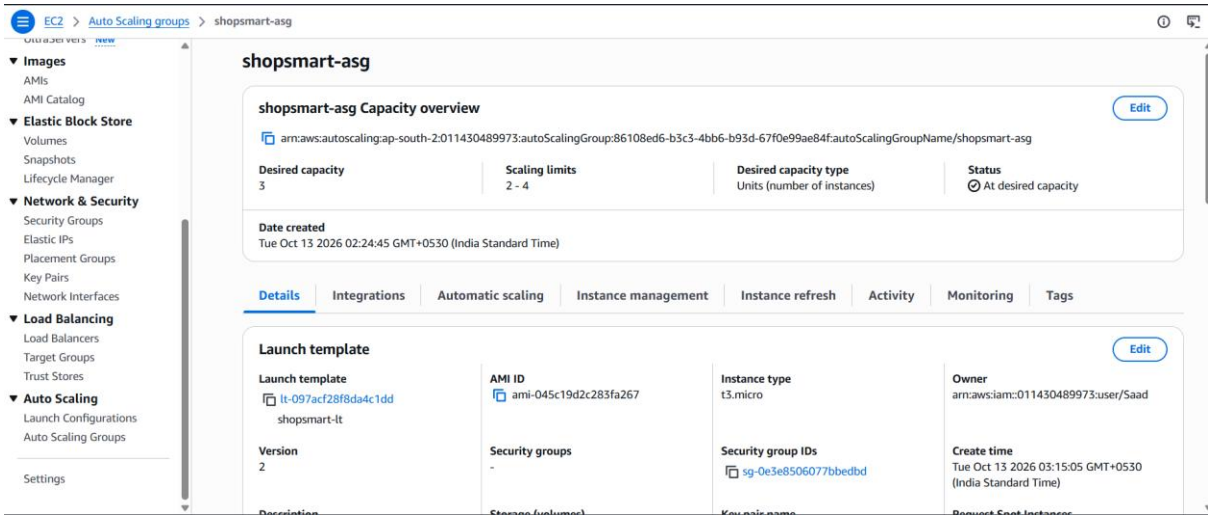
4.1 Launch Template

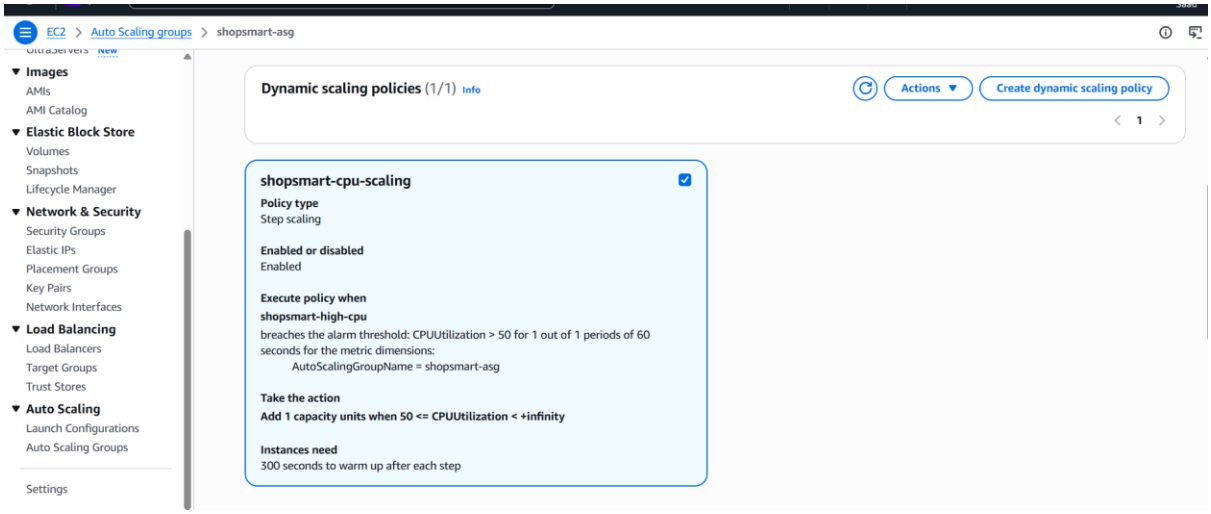
The shopsmart-lt launch template defines the configuration used to launch EC2 instances for the ShopSmart application. It references the custom AMI shopsmart-ami-v1 and specifies the instance configuration, security settings, and storage options.



4.2 Auto Scaling Policy

The shopsmart-asg Auto Scaling Group manages the application's EC2 capacity. The shopsmart-cpu-scaling policy is configured to respond to CPU utilization according to its associated CloudWatch alarm and scaling adjustment. This enables the infrastructure to add capacity when demand increases.

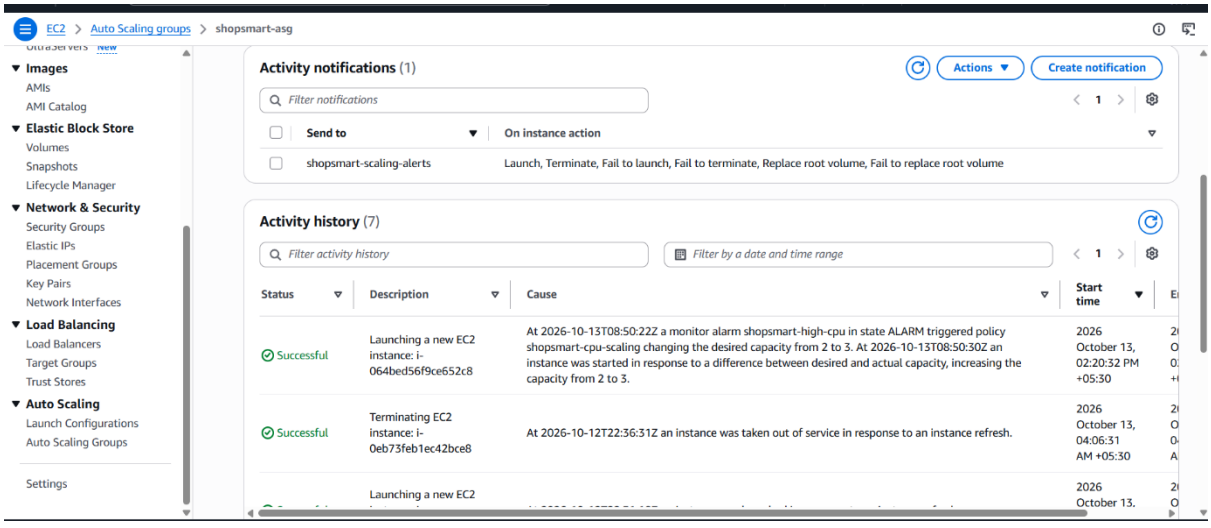




5: Successful Scale-Out and Unhealthy-Instance Replacement

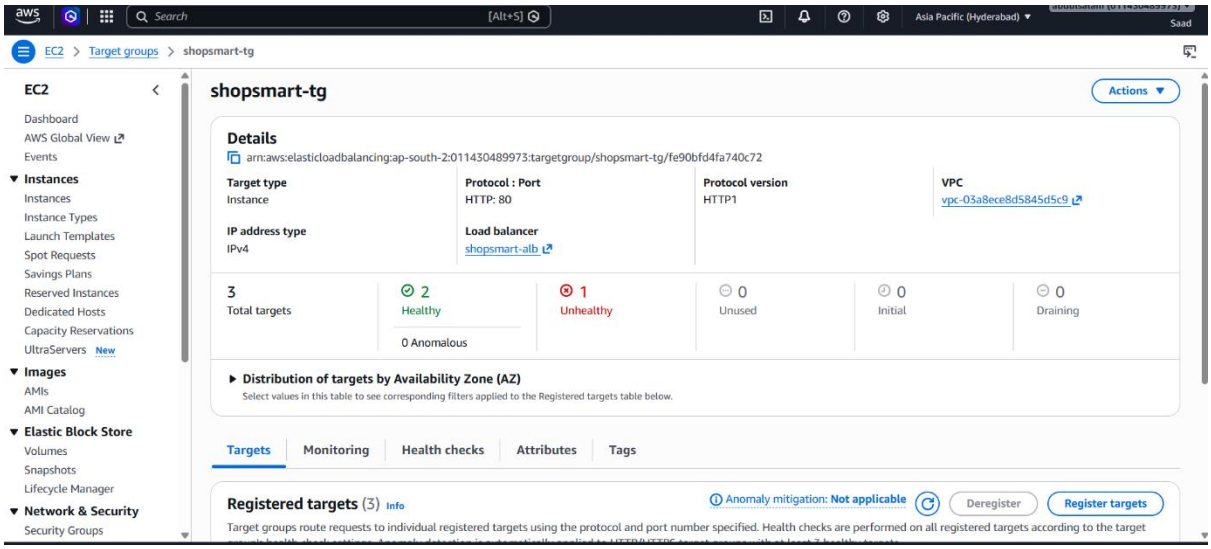
5.1 Successful Scale-Out

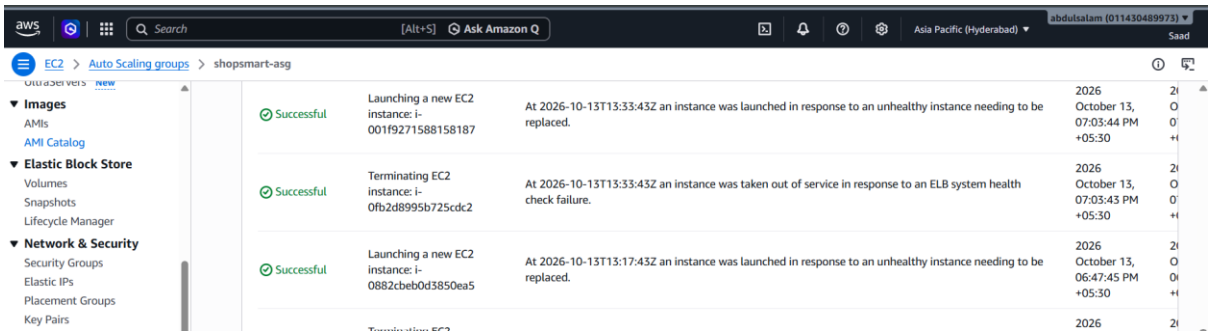
The ShopSmart Auto Scaling Group is configured to increase capacity when the associated CloudWatch CPU alarm breaches its threshold. When CPU utilization exceeded the configured threshold, the scaling policy initiated the launch of an additional EC2 instance. The successful launch event in the Auto Scaling activity history provides evidence of scale-out.



5.2 Unhealthy-Instance Replacement

The Auto Scaling Group can replace an instance that is identified as unhealthy, provided the relevant health checks and replacement behavior are configured. The activity history should be used to document the actual replacement event and its outcome.



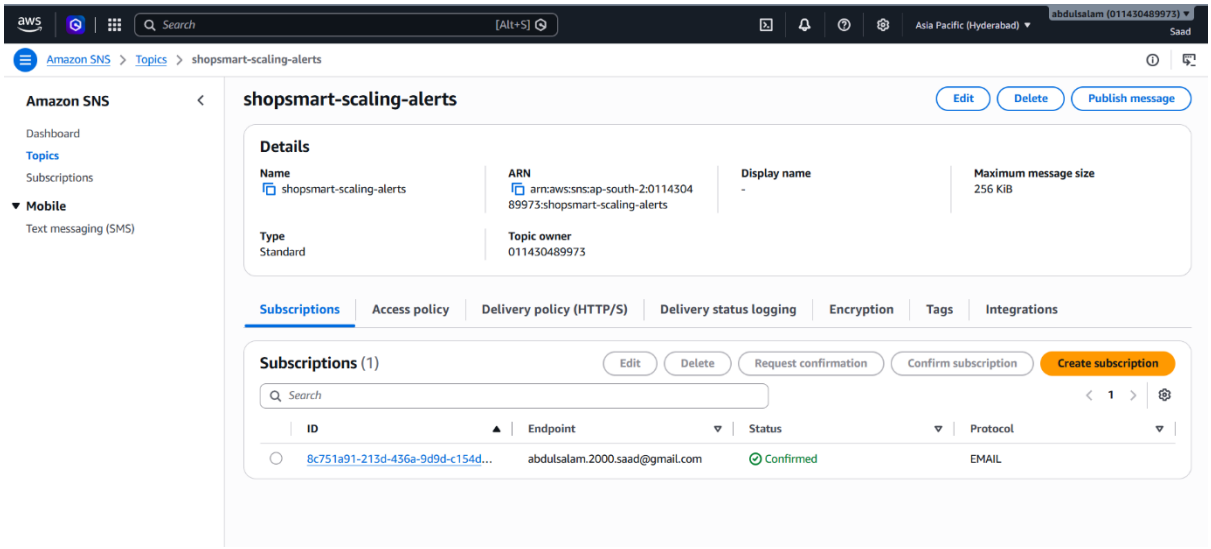


6: Amazon SNS Notifications

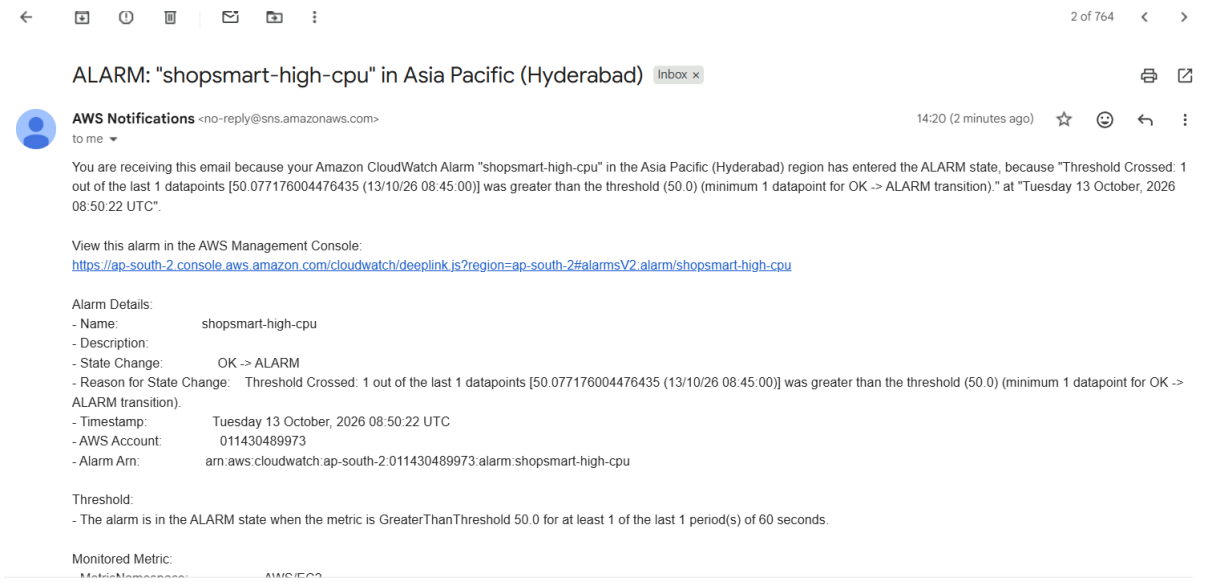
Amazon Simple Notification Service (SNS) is configured to support notifications for the ShopSmart Auto Scaling Group. The SNS topic shopsmart-scaling-alerts is associated with Auto Scaling activity notifications for instance launches, terminations, and selected failure events.

The email subscription allows notifications to be delivered to the subscribed email address. These notifications help the administrator monitor scaling activity and identify infrastructure events.

SNS topic with confirmed email subscription.



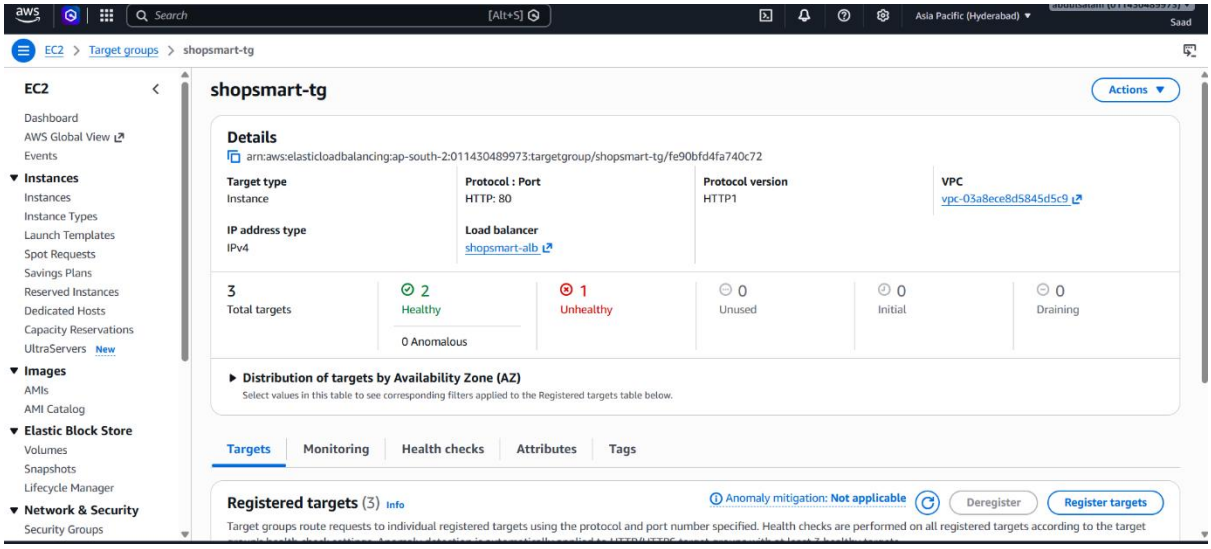
Email notification for the shopsmart-high-cpu alarm.



7: Incident Report

Incident Description

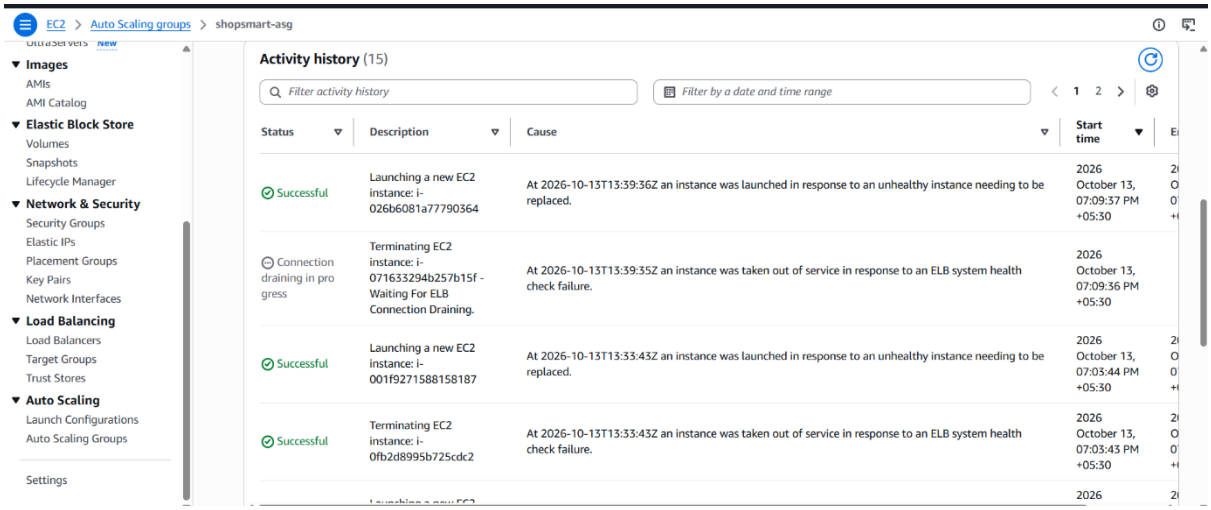
During testing of the ShopSmart application, Nginx was temporarily stopped on one EC2 instance to simulate a web server failure. As a result, the Application Load Balancer's health checks failed, and the instance was marked as unhealthy.



Resolution

The Auto Scaling Group was configured to use EC2 and Elastic Load Balancing health checks. After the unhealthy instance was detected, Auto Scaling initiated its termination and launched a replacement instance using the configured launch template and AMI.

Nginx was restored or initialized on the replacement instance through the launch configuration, and the instance could then serve application traffic after passing its health checks.



8: Cost-Optimization Proposal

Objective

The objective is to reduce the operating cost of the ShopSmart web application while maintaining high availability, security, and the ability to handle traffic spikes.

Comparison of EC2 Purchasing Models

Purchasing model	Cost and benefit	Suitability for ShopSmart
On-Demand Instances	Pay for compute capacity by usage, without a long-term commitment.	Suitable for development, testing, and temporary scaling capacity.
Savings Plans	Offer lower compute rates in exchange for a commitment to a consistent amount of eligible compute usage.	Suitable for the predictable baseline workload after usage patterns are understood.
Reserved Instances	Provide discounted pricing for eligible EC2 usage in exchange for a commitment, subject to the selected terms.	Worth evaluating if the production workload remains stable over a long period.
Spot Instances	Use spare AWS capacity at potentially substantial discounts, but instances can be interrupted.	Suitable for fault-tolerant background jobs, not as the only capacity serving the main website.

Recommended Cost-Optimization Strategy

1. Maintain the Auto Scaling Group's minimum capacity of two EC2 instances across two Availability Zones to support availability.

2. Allow the Auto Scaling Group to launch additional instances during high traffic and remove excess capacity when demand falls.
3. Use On-Demand Instances initially because the application's long-term compute usage is not yet established.
4. Consider a Savings Plan for predictable baseline usage after reviewing actual utilization and projected demand.
5. Use Spot Instances only for suitable interruptible background tasks, if such workloads are introduced.
6. Reduce unnecessary storage costs by reviewing EBS volumes and snapshots and deleting only resources that are no longer required.
7. Review CloudWatch metrics and AWS billing regularly to identify idle resources and unexpected charges.